Three Validations of the Solver

Field-line bending · droplet elongation · Bacri Fig. 3

MTP — Negative rotational viscosity in ferrofluids via phase-field MRT-LBM

Why three, and why these three

A solver is only as trustworthy as the tests it survives. We chose three that each stress a *different* part of the machinery:

- **Field-line bending** checks the *magnetostatics alone* — does the h-lattice compute the right field around a magnetizable body? (Flow frozen.)
- **Droplet elongation** checks the *coupling of field, interface, and flow* — does a magnetic force deform a two-phase interface correctly against surface tension?
- **Bacri Fig. 3** checks the *full dynamic magnetization* — does the MRSh law reproduce the negative-viscosity effect, the whole point of the project?

Passing all three means the field solver, the two-phase coupling, and the magnetization dynamics are each independently sound.

1 Validation 1 — Bending of magnetic field lines (cylinder in a uniform field)

1.1 What it tests

A circular magnetizable cylinder sits in a uniform applied field. A more permeable body *pulls field lines into itself* — they bend inward and the interior field is uniform but weaker than applied. This is pure magnetostatics: it isolates the h-lattice from all flow and magnetization dynamics.

The physics, plainly

Field lines are "lazy" — they prefer to travel through the easiest material. A high-permeability cylinder is an easy path, so lines crowd *into* it and bend to enter. Inside, the field is uniform because the geometry is symmetric; its strength is set only by the permeability ratio. The analytic answer for a 2D cylinder is

$$\frac{H_{\text{in}}}{H_0} = \frac{2}{\mu_1/\mu_2 + 1},$$

(Paper Eq. 53). If the lattice reproduces this number, the field solver is correct.

1.2 Setup and reduction

The reference paper (Hu, Li & Niu, *PRE* **98**, 033301, 2018) uses a linear $\mathbf{M} = \chi\mathbf{H}$ law. Our solver uses the nonlinear Langevin/MRSh law, so we drive it into its *linear limit* $\xi_e \ll 1$, where

$$\chi_{\text{eff}} = \tfrac{1}{3}\phi M_{\text{s}}\,\xi_{\text{per }H}, \qquad \mu = \mu_0(1+\chi).$$

A per-node map assigns χ_1 inside the cylinder and χ_2 outside (via $M_{\text{s}}^{\text{local}} = 3\chi/\xi_{\text{per }H}$, $\phi \equiv 1$ throughout), sweeping $\mu_1/\mu_2 \in [1, 4]$.

1.3 Boundary conditions

Domain	200×200 lattice, cylinder radius $R = 20$ at centre.
Flow	**Frozen** (`FREEZE_FLOW=true`) — pure magnetostatics, no f/g update.
Applied field	Uniform, static, $+x$: $H_d = 0.005$ (linear regime, $\xi_e \approx 0.058$).
h-lattice walls	Neumann on A_z carrying the uniform applied field, $\partial A_z/\partial n = \mu_0(H_{\text{app}} + M_w) \cdot \hat{\mathbf{t}}$.
Cylinder "surface"	No explicit interface BC — continuity of normal **B** and tangential **H** is *automatic* in the A_z formulation, since A_z and its derivatives are continuous.

Interior field matched Paper Eq. 53 with 0–3.8% error across the permeability sweep. The residual was diagnosed as **finite-domain far-field truncation** (the uniform field is only truly uniform at infinity), *not* a solver error — confirmed by checking that the interior field was spatially uniform.

2 Validation 2 — Ferrofluid droplet elongation

2.1 What it tests

A circular ferrofluid droplet in a uniform field stretches into an ellipse: the magnetic (Maxwell/Kelvin) stress at the interface pulls it along the field, surface tension resists, and it settles at an aspect ratio b/a. Unlike Validation 1, the observable *is* a flow result — the equilibrium shape emerges from moving the interface.

Why the flow cannot be frozen here

In the cylinder test the field alone was the answer, so we froze the flow. Here the answer is a *shape*, and shape only changes if fluid *moves*. The magnetic force deforms the interface, surface tension pulls back, the droplet oscillates and relaxes to an ellipse. Freeze the flow and nothing deforms. So `FREEZE_FLOW=false`, and this test exercises the g, f, and h lattices together.

2.2 Setup and reduction

Droplet radius $R = 12.5$, susceptibility $\chi_0 = 2.2$. The mapping $M_s\,\xi_{\text{per }H} = 3\chi_0 = 6.6$ makes $\chi_{\text{eff}} = \chi_0$ inside the droplet ($\phi = 1$) and 0 outside ($\phi = 0$) *automatically* through the ϕ factor — no geometric map needed. The controlling group is the magnetic Bond number $\text{Bo}_m = \chi_0^2 H^2 R/\sigma$; deformation follows $b/a - 1 \approx k\,\text{Bo}_m$ at small fields, becoming convex as demagnetization feedback accelerates elongation.

2.3 Boundary conditions

Domain	≈ 200 wide box, droplet centred; periodic in x.
Flow	**On**. No-slip $u = v = 0$ on the channel walls.
Applied field	Uniform, static. Paper applies $+y$; we apply $+x$ (deformation is isotropic, so b/a is identical — the droplet simply elongates along x).
Field side walls	$\partial H/\partial n = 0$ (zero normal field gradient), i.e. Neumann on A_z.
Interface	Captured by the phase field ϕ (Allen–Cahn); the $\phi = 0.5$ contour defines the droplet; b/a read with sub-pixel interpolation at equilibrium ($\max\|\mathbf{u}\| \to 0$).

At large fields ($b/a \gtrsim 4$) the droplet becomes a filament longer than the box, and the **periodic-x boundary lets it wrap and self-interact**. So the low-field points calibrate the field scale (`SCALE`≈ 0.006, cross-checked by geometric mean of the bracketing values), and the solver's own nonlinearity carries the high-field convexity — which correctly reproduces the paper's convex $b/a(H)$ curve. The top two points are unreliable until the domain is enlarged or the walls are made non-periodic; this is a domain-sizing issue, not a physics error.

3 Validation 3 — Bacri Fig. 3 (the negative-viscosity effect)

3.1 What it tests

This is the project's central target: a Poiseuille flow under an *oscillating* field, where the AC field *lowers* the effective viscosity — and at strong enough coupling drives it *negative*. It tests the full MRSh magnetization dynamics, the flow–field coupling, and the AC sub-cycling machinery together.

How a field can lower viscosity

Normally, a field resists particle rotation and *raises* viscosity (a static field always does this). But an *oscillating* field is different. The particle magnetization lags the field. If the field oscillates near the relaxation rate, the lagging magnetic torque can end up pointing the *same* way the flow is already trying to spin the particles — so instead of fighting the flow it *helps* it. The measured quantity is the reduced torque coefficient g; when the field-driven torque adds to the flow, $g < 0$ and the effective viscosity drops below its zero-field value. Reproducing Bacri's $g(\xi_0, \omega\tau_B)$ curve is the proof.

3.2 Setup

Body-force-driven 2D Poiseuille flow (uniform $\mathbf{F}_b = 2 \times 10^{-7}\,\hat{x}$ standing in for a periodic pressure gradient), channel height $H = 50$, with a linearly polarized oscillating field $\mathbf{H}_0(t) = H_0 \cos(\omega t)\,\hat{x}$ parallel to walls and flow. Effective viscosity extracted from the converged velocity profile; g extracted from a harmonic fit over whole field periods.

3.3 Boundary conditions

Domain	$n_X \times n_Y$ channel, $H = 50$; periodic in x.
f, g **walls**	Full bounce-back (halfway no-slip at $y = -\frac{1}{2}$ and $y = n_Y - \frac{1}{2}$); ghost $u_{-1} = -u_0$.
h **walls**	Specular reflection + flux injection carrying $\partial A_z/\partial n = \pm\mu_0(H_{0x} + M_{x,w})$, with $M_{x,w} = \frac{3}{2}M_{x,0} - \frac{1}{2}M_{x,1}$.
Field time-dependence	Oscillating: applied field scaled by $\cos\omega t$; h-lattice **sub-cycled** ($N_{\text{sub}} = 64$) each physical step so A_z tracks the drive within the causality floor.
ϕ **ghost**	Mirrored $\phi_{-1} = \phi_1$ (single-phase here, $\phi \equiv 1$).

Static-field agreement $\sim 0.3\%$ across all ξ_0. AC agreement $\sim 1.6\%$ up to $\xi_0 = 10$ (at $\omega\tau_B = 2$), rising to $\sim 4.5\%$ at $\xi_0 = 20$. **Fig. 3 of Bacri et al. was successfully reproduced** with the body-force-driven solver. The residual at high field under AC is not yet fully attributed — a known open item.

What made the agreement this tight

Four fixes moved the numbers from "roughly right" to $\sim 1\%$: (1) interpolating **H** across RK4 sub-steps instead of freezing it (removed up to 34% bias); (2) resolving the *reference* theory curve properly, since its own RK4 was unstable when $h\,\xi_0^2 > 2.785$ (removed up to 15% inflation); (3) computing η_0 from converged, not transient, snapshots (7.6% bias); (4) fitting g over whole field periods. Each was verified against an independent integrator.